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# Mode-mixity and dynamic failure mode transitions in polycarbonate

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**Abstract**

The combination and transition of a shear to opening type of failure mechanism has been reported by for *dominant mode II loading* of notched or cracked plates (metallic alloys and polycarbonate). The present paper addresses additional aspects of the phenomenon in relation to *mode-mixity* for actual fatigue cracks in polycarbonate specimens. Two distinct experimental setups are used and systematically compared throughout the work, by means of numerical simulations and scanning electron fractographic analysis. The first setup allows for dominant mode II loading with a minor mode I component (side impact of cracked plates). A preliminary numerical study accounting for contact and friction between the fatigue crack flanks is carried out. This study shows that, due to the fatigue crack, the initial loading mode is close to "pure mode II" followed at a later stage by mixed-mode. The experiments show that by varying the impact velocity, adiabatic shear band formation and fracture can be provoked and controlled, as observed in previous work. For these experiments, a threshold value of $K_{II}$ (4 MPa $m^{1/2}$) is proposed to trigger shear band formation in the adiabatically heated crack-tip material. The second setup allows for dominant mode I loading with a minor mode II component, using compact compression specimens (CCS). Here, mixed-mode is experienced from the onset of the crack-tip loading throughout the experiment. The comparative fractographic analysis shows that identical characteristic failure mechanisms operate irrespective of the specimen geometry (i.e. *mode-mix*) and crack-tip nature (notch or crack) for a similar range of impact velocities. Specifically, a shear failure mechanism is *also* observed in those "mode I" (CCS) specimens which were impacted at a higher velocity. This comparative study therefore extends the phenomenon of failure mode transition to general mixed-mode loading. © 1998 Elsevier Science Ltd. All rights reserved.

*Keywords:* Dynamic fracture; Failure mode transitions; Mode-mixity; Polymeric materials; Finite elements; Fractography

## 1. Introduction

Crack initiation under transient dynamic loading is an important topic as it is likely to occur in many structures which are not always intended or designed for that purpose. Whereas the framework of quasi-static fracture mechanics is fairly well established both theoretically and experimentally (see e.g., Tetelman and McEvily, 1967; Kanninen and Popelar, 1985; Hertzberg, 1989), less is known about dynamic fracture. Considering the firm theoretical basis developed for linear elastic materials (Freund, 1990), it appears by contrast that many problems are still unsolved due to a lack of experimental evidence. One possible reason may be the typical short time scales involved (microseconds) together with the need for sophisticated experimental apparatus. Several years ago, Knauss

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(1983) compared the body of evidence concerning two phases of the dynamic fracture process, initiation and propagation. His conclusion was that less is known about initiation than propagation. As of today, there is no commonly accepted criterion for dynamic crack initiation by contrast with the case of static loading.

Between elastic (small scale yielding) and fully plastic (general yield) fracture comes the interesting case of failure mode transitions according to the loading conditions. Considering dynamic fracture, such transitions were reported for commonly used engineering alloys in which the failure mechanism(s) depends on the impact velocity. This has long been known for the case of Charpy tests for instance (ductile to brittle transition for crack opening mode). A new kind of failure mode transition has been reported following side impact of notched (and cracked) plates. In these experiments, a projectile is shot parallel to the notch direction, inducing mixed-mode loading conditions with a dominant mode II term. Kalthoff (1988) investigated maraging steel and reported that at lower impact velocities, fracture proceeds in a "brittle" way by forming an angle of about 70° with respect to the initial notch line. Increasing the impact velocity causes a shear band to develop and propagate for a short distance almost along the continuation of the original notch line. Lee and Freund (1990) studied this problem and provided an analytical solution for short times after impact such as to exclude wave reflections as a result of the specimen's finite size. They showed the development of a minor negative mode I component aside the dominant mode II. Mason et al. (1992) used full field optical methods to monitor the crack-tip in this setup. They found an excellent agreement with Lee and Freund's solution while extending it to longer times. Mason et al. (1994) investigated the deformation field around an adiabatic shear band in maraging steel plates. They modeled the band as a mode II Dugdale zone and got an estimate of the value of $K_{II}$ at which the shear band initiates. Ravi-Chandar (1995) investigated polycarbonate and he reported the very same behavior also in terms of a brittle to ductile transition and fracture angle (66°). Zhou et al. (1996a, b) carried out a detailed investigation of side impact in maraging steel and a titanium alloy. They report a shear to opening (with an angle of about 30°) fracture mode transition in the maraging steel. The titanium specimens fail by shear mechanism only. These authors also carried out a finite element analysis with thermo-mechanical coupling to account for the generation and growth of the shear band. Their study showed that a significant fraction of the melting temperature can be reached in the band, according to the investigated material.

The cited works report about systematic shear band generation at the higher impact velocities with varying amounts of propagation and subsequent failure by opening mechanism at specific angles (either 70° or 30°). These works address *side impact experiments only* with the crack-tip fields determined by using optical methods (caustics or CGS). They also outline the influence of adiabatic crack-tip heating in triggering a shear mode of failure.

*It is therefore natural to wonder whether such observations and related failure mechanisms are restricted to side impact configurations, or more specifically how do they fit in the general framework of mixed-mode loading.*

Consequently, this paper addresses dynamic crack initiation subjected to mixed-mode loading. The selected material is commercial polycarbonate. We investigate actual cracks, grown experimentally as fatigue cracks and modeled numerically through contact and friction conditions. To deal with mode-mixity *we systematically investigate and compare* situations in which mode II is *either* the dominant (shear impact) *or* the minor mode (compact compression specimens, Rittel et al., 1992). Failure modes are characterized using optical and scanning electron microscopy. The results are discussed and compared such as to outline the role of mode mixity in this kind of experiments in particular and in dynamic crack initiation more generally.

The paper is organized as follows: firstly we present the experimental setup, data analysis and simulation framework. Next, we present numerical results on the crack-tip fields for actual fatigue cracks typical of those investigated experimentally. In Section 3 we report, compare and discuss the

results obtained in the two types of experiments. Section 4 summarizes several important points of this work, followed by concluding remarks.

## 2. Experimental and numerical framework

### 2.1. Experimental

The experimental setup comprises the impacting apparatus and the specific specimen (Fig. 1). We use an 12.7 mm diameter instrumented (PH 17-4) steel bar, which on one side is in contact with the specimen and on the its other side is impacted by a striker fired by a gas gun. Typical impact velocities range from 8 to 60 m/s. Strikers with length of 8 or 17 cm were used to produce impacts of varying duration (30 and 64 μs, respectively). The striker induces a compressive pulse ($\varepsilon_{in}$) in the bar which is partly transmitted to the specimen and partly reflected as a tensile pulse ($\varepsilon_{ref}$). The pulses were collected by means of a digital oscilloscope (Nicollet 490 model) at a sampling rate of 5 MHz. The experimental pulses were corrected for geometrical dispersion (Kolsky, 1963; Lifshitz and Leber, 1994) using home made programs. The interfacial force $P_i$ and velocity $V_i$ pulses were determined from the incident and reflected pulses (Kolsky, 1963).

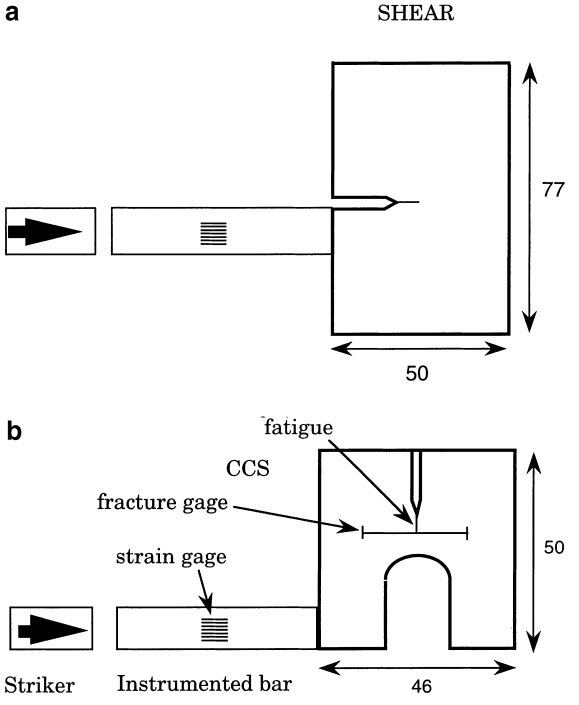



Fig. 1. Experimental setup and specimen dimensions (in mm). (a) Side impact. (b) Compact compression specimen.

Two kinds of specimens were used: the rectangular plate, for shear impact ("mode II") experiments and the compact compression specimen (CCS) for "mode I" experiments. Due to the asymmetrical loading, the crack-tip experiences mixed-mode conditions: dominant mode for the CCS (Maigre and Rittel, 1993) and dominant mode II for the cracked plate (Kalthoff, 1988). Throughout the experiments, the specimen was brought in contact with the instrumented bar without special fixtures to support it so that fracture resulted from inertia only (see also Kalthoff et al., 1983; Giovanola, 1986).

In each specimen, a fatigue crack was grown carefully such as to minimize the damage which accumulates at the crack-tip. The typical fatigue extension was about 2–3 mm long. Fatigue cracks were modeled numerically as described in the next section.

The investigated material was polycarbonate, supplied as 12.7 mm thick sheets. The mechanical properties of this material were assessed using quasi-static tensile testing (Table 1). An equivalent "transfer modulus" (to account for dynamic effects – see Rittel and Maigre, 1996a) was determined from longitudinal wave velocity measurements using ultrasonic technique. We also measured the quasi-static fracture toughness of the material using fatigue precracked CCS's (stroke control at a velocity of 0.2 mm/min). A representative value was found to be $K_{IC} = 3.6$ MPa $m^{1/2}$.

Determination of the onset of crack-propagation (fracture time) was achieved using single

Table 1
Measured mechanical properties of the commercial polycarbonate used in this study

| | Young's modulus (GPa) | Poisson's ratio |
|---|---|---|
| Static | 2.43 | 0.37 |
| Dynamic | 3.72 | 0.37 |

(timing) wire fracture gages with the CCS specimens which fail by separation in two pieces. In those side impact experiments where cracks actually grew, propagation was limited to about 2 cm before the crack arrested in the specimen. As a result very little opening ensued which would fracture the timing wire. Consequently, this technique was of limited applicability and fracture time was not directly measured in the side impact experiments.

### 2.2. The numerical model

Throughout this study we adopt the hybrid experimental–numerical approach (Kobayashi, 1987) by which additional aspects of the experiment can be highlighted through proper numerical simulation. However, the numerical simulations are kept *as simple as possible* such as to illustrate relevant aspects of the experiments with a minimum number of adjustable parameters.

#### 2.2.1. Basic assumptions

The range of validity of a single parameter crack-tip characterization has been extensively discussed for (impact loaded) stationary and running cracks (see e.g., Mason et al. 1992; Taudou et al., 1992). We investigate impact loaded *stationary cracks* for which the crack-tip fields can be described by the stress intensity factors $K_I$ and $K_{II}$, as shown by Mason et al. (1992). We assume small scale yielding where adiabatic thermal phenomena remain confined to the plastic zone surrounding the crack-tip. We did not include in this study thermo-mechanical couplings as in Zhou et al. (1996b) since the emphasis here is on the analysis of the crack-tip fields prior to initiation without attempting to model accurately shear band growth or phenomena related to the process zone.

The stress intensity factors are determined from simple numerical (simulations) models of actual experiments and these results can thus serve as a basis for the analysis and comparison of the various experiments reported below.

#### 2.2.2. Modeling

Each kind of specimen was analyzed using a commercial finite element code (ANSYS, 1994). We assumed two-dimensional plane strain deformations and linear elastic material behavior. The crack-tip is modeled using singular quarter point elements (Barsoum, 1978) and the crack in all the analyses is kept stationary so that the stress intensity factors are a function of time only (Freund, 1990). Traction free conditions are assumed over all the specimen except at the specimen–bar interface. To simulate the experiment, the recorded force pulse ($P_i(t)$) is applied to this interface as boundary condition. Once the specimen–bar separation has occurred, measurement of the interfacial forces and velocities using the instrumented bar is not possible as the specimen is no longer loaded through the input bar. However, the crack-tip fields can be assessed for any duration by extending the calculation while applying traction free conditions.

Fatigue cracks induce two specific conditions. The first is contact between the crack-flanks to avoid interpenetration following the negative mode I which develops for shear impact experiments (Lee and Freund, 1990). Secondly, friction may develop between the crack faces which affects the mode II component of the crack-tip loading. Both the contact and/or friction conditions induce a nonlinearity which precludes the use of the path independent H-integral (or other kinds of dynamic weight functions) as well as the use of superposition.

Contact was modeled using special contact elements (CONTAC48) which apply a normal reaction when the gap between the two surfaces is closed. Friction was modeled as (elastic) Coulomb friction and the coefficient of friction in all the analyses was chosen as $\mu$'s ($=0.4$). This value is purely arbitrary since the actual coefficient of friction was not measured but it provides a basis for comparison, rather than for strictly quantitative estimation. For closer replication of the experimental conditions, contact was prescribed on the fatigue crack only. The equations of equilibrium were solved using Newmark implicit scheme (Bathe, 1982). The mode I and mode II stress intensity factors were determined from the displacements "measured" at two control points in the vicinity of the crack-tip (Bui et al., 1992).

### 2.3. A preliminary study of impact loaded fatigue cracks (mode II)

A preliminary simulation was carried out to assess the influence of friction and contact on the stress intensity factors in our specimens. Here a gaussian pressure pulse (30 μs wide – $2 \times 10^7$ Pa at peak) was applied to simulate transient loading characteristic of that applied in typical experiments. For the sake of brevity, we report only results pertaining to the side impact experiments.

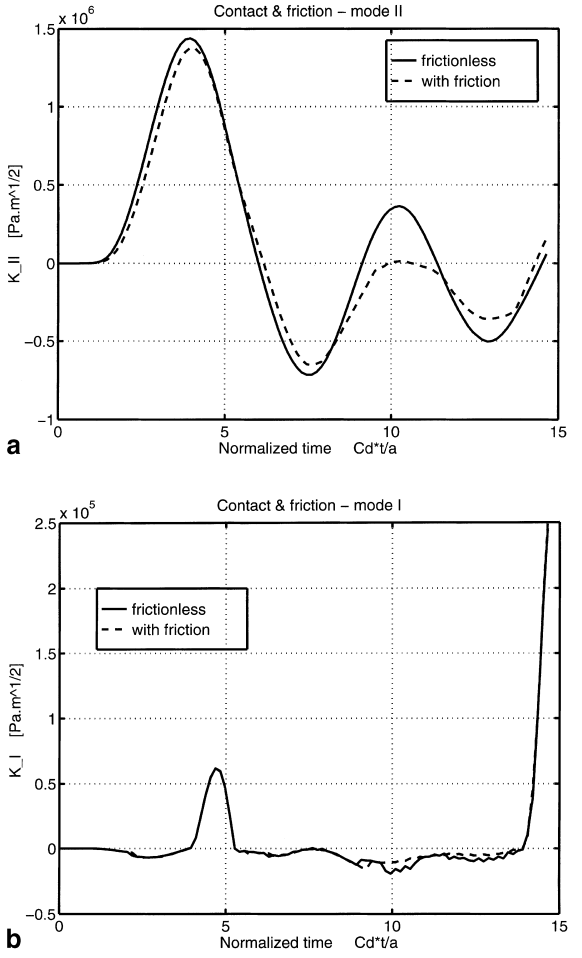



Fig. 2. Numerical simulation of side impact on a fatigue crack with prescribed contact w/o frictional conditions between the crack flanks. Evolutions of the stress intensity factors. (a) Mode II and (b) mode I as a function of normalized time Cd·$t/a$. (Cd and $a$ stand for longitudinal wave speed and crack length.)

In Fig. 2 are shown the evolutions of $K_I$ and $K_{II}$ (the values bear no physical meaning) for the same shear impact specimen to which we prescribed contact conditions with $\mu = 0$ (frictionless) and $\mu = 0.4$. The time scale has been normalized to take into account the crack length and wave velocity, as in the analytical results of Lee and Freund (1990). Our "numerical experiments" with frictional cracks show, as expected, that contact does not allow for the negative mode I to develop while friction does not seem to affect crack opening (the slightly negative $K_I$ values are most likely related to the contact stiffness used in the numerical calculations). As time increases ($t_{norm} > 10$) a marked tendency for crack opening develops. On the other hand, the influence of friction is felt on the mode II in the sense that contact tends to restrain the shearing motion of the crack faces, but this is mostly evidenced at the later times ($t_{norm} > 10$). By comparison with previously reported results (e.g. the experimental extension of Lee and Freund's results to longer times by Mason et al., (1992), it can be noted that here too there is a transition from shear to opening mode as time increases. However, the absence of a negative mode I component during a large initial time brings this experiment close to a "pure mode II" condition. While being expected intuitively, this result is quite important as it can immediately be applied to *dynamic mode II testing* without devising complicated experimental setups. The introduction of fatigue cracks provides the necessary boundary conditions to achieve this goal.

## 3. Results

### 3.1. Dominant mode II (Shear experiments)

Shear impact experiments of two kinds were carried out. The first kind is identical to the other experiments reported in the literature (side impact of a free plate). The other kind of experiments involved constraining of the specimen's upper and lower faces (subsequently referred to as confinement) to cancel or at least minimize the crack opening mode (Fig. 3). These experiments were carried out to isolate the influence of the opening

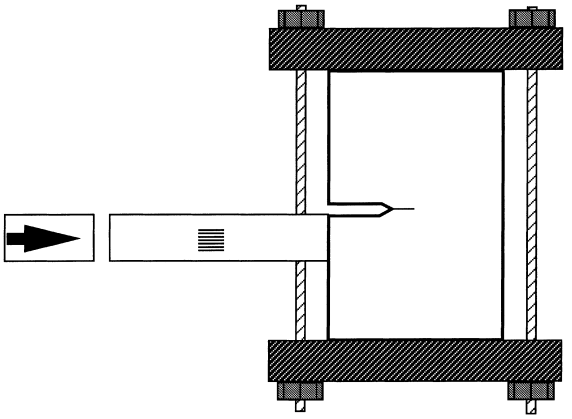


Fig. 3. Schematic representation of a side impact specimen with applied confinement to restrain crack opening.

mode on the failure process. The initial angle (kink angle) at which the crack grew with respect to the initial crack line was measured on both faces of the specimen (accurate determination of the kink angle is sometimes delicate for continuously curving cracks). We also observed that the fatigue cracks did not always grow uniformly through the thickness. Therefore, in all calculations, the average crack length was used as a further simplification of the model. In some experiments, the impact velocity is missing as a result of experimental problems. A total of 16 specimens were tested as listed in Table 2.

As a first remark it can be noted that in these experiments too there is a range of velocities (the

Table 2
Summary of the experimental results for shear impact specimens

| Sample # | $v_{imp}$ (m/s) | Crack ($\times 10^{-3}$m) | Fatigue Y/N | Kink angle (°) | Remarks |
|---|---|---|---|---|---|
| 1 | n.a. | 15.0–15.3 | Y | 41.8 | |
| 2 | 27 | 15.0–15.2 | Y | 37.6 | FEM |
| 3 | 35 | 13.2–17.4 | Y | 37.2 | |
| 4 | 35 | 15.6–17.6 | Y | 40.7 | |
| 5 | 25 | 13.4–17.0 | Y | 36.2–43.5 | |
| 6 | n.a. | 15.9–16.2 | Y | ≈0 | SB |
| 7 | 12 | 14.4–15.0 | Y | 30.0 | |
| 8 | n.a | 15.6–16.7 | Y | ≈0 | SB |
| 9-1 | 15 | 13.8–14.9 | Y | no fracture | 1st impact |
| 9-2 | 18 | 13.8–14.9 | Y | no fracture | 2nd impact |
| 9-3 | 22 | 13.8–14.9 | Y | 37.2 | 3d impact |
| 10 | n.a. | 15.2–15.9 | Y | ≈0 | |
| 11 | 37 | 12.9–12.9 | N | 36.5 | |
| 12 | 60 | 16.3–17.2 | Y | ≈0 | SB-FEM |
| 13-1 | 28 | 12.8–13.0 | N | no fracture | SB-1st impact-confined |
| 13-2 | 35 | 12.8–13.0 | N | no fracture | 2nd impact-confined |
| 13-3 | 35 | 12.8–13.0 | N | 34.0–40.0 | 3d impact-free |
| 14-1 | 32.5 | 16–17.5 | Y | no fracture | SB-1st imp.-confined-FEM |
| 14-2 | 32.5 | 16–17.5 | Y | no fracture | 2nd impact-confined |
| 14-3 | 32.5 | 16–17.5 | Y | no fracture | 3d impact-confined |
| 14-4 | 32.5 | 16–17.5 | Y | 37.5–38.0 | 4th impact-free-FEM |
| 19 | ≈60 | 16.5–17.8 | | no fracture | SB |
| 23-1 | 32.5 | 16.0–16.5 | Y | no fracture | SB-1st impact-confined |
| 23-2 | 35 | 16.0–16.5 | Y | no fracture | 2nd impact-confined |
| 23-3 | 24 | 16.0–16.5 | Y | no fracture | 3d mpact-confined |
| 23-4 | 24 | 16.0–16.5 | Y | no fracture | 4th impact-confined |
| 23-5 | 24 | 16.0–16.5 | Y | no fracture | 5th impact-confined |
| 23-6 | 24 | 16.0–16.5 | Y | no fracture | 6th impact-confined |
| 23-7 | 22.5 | 16.0–16.5 | Y | no fracture | 7th impact-free |
| 23-8 | 22.5 | 16.0–16.5 | Y | 28.0–29.5 | 8th impact-free |

The table indicates impact velocity, crack length as measured on both sides of the specimen, fatigue precracking (Y/N) and the measured kink angle. In some instances, the crack did not propagate as such but a shear band was observed. The remarks column details the number of impacts, the use of confinement. FEM stands for numerical modeling of this experiment, and SB, shear band when detectable by visual examination only.

"lower impact velocities", e.g., below 25 m/s) for which the specimen fractures whereas in another range (the "higher impact velocities", e.g., 60 m/s) an adiabatic shear band is observed and the crack does not seem to grow (Ravi-Chandar, 1995; Rittel et al., 1997). Shear bands are clearly detected by visual examination only on specimens subjected to the high impact velocities (Table 2). The adiabatic shear band is typically 1–2 mm long (Fig. 4) and it has the planar appearance which was reported by Ravi-Chandar (1995), although it is shorter than the reported 10 mm.

All specimens, but #11 and #13, contained fatigue cracks. It can also be noted that regardless of the kind of the crack-tip, all the specimens which fractured (11 out of 16) did it at angles which did not exceed 40°. This value must be contrasted with the predicted value of 70° at which maximum tensile stress develops for mode II loading. Consequently, opening at such an angle suggests the joint operation of not only mode II but also mode I fracture mechanism as discussed later.

Keeping in mind the development of the positive mode I component at later times, additional light can be shed on its role by considering the confined shear impact experiments (#14 and 23). Confinement reduces significantly the mode I component so that (final) kinked fracture did not occur as observed in other specimens. However, shear bands were observed suggesting that shear band formation is not affected by the boundary conditions of this experiment. When the confinement was released, the shear band either grew further or final fracture was observed at a characteristic angle according to the impact velocity. By contrast, it will be mentioned that we carried out the same confined experiments on fatigue precracked brittle polymethylmethacrylate (PMMA) specimens. For high velocity impacts, we always observed fracture (and even shattering) at 70° (or greater) irrespective of the presence or lack of confinement.

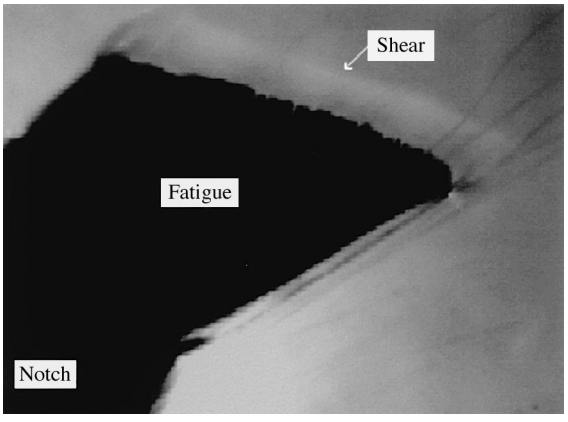



Fig. 4. Shear band viewed by transparency in a tilted side impacted specimen. Optical micrograph. The shear band is planar and extends across the specimen along the original crack line.

### 3.2. Numerical simulation of selected shear impact experiments

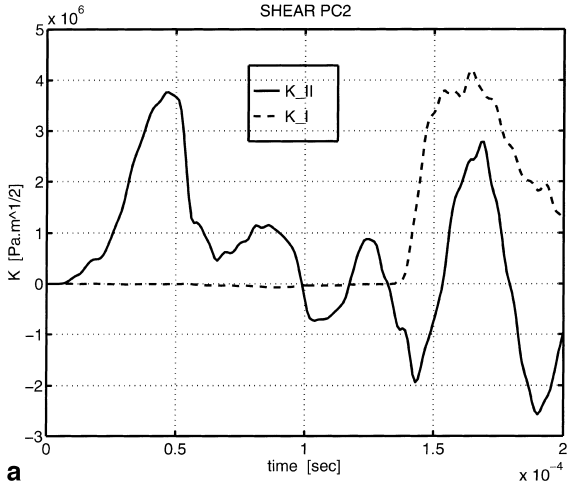
We will present and discuss results obtained for the numerical simulations of two characteristic experimental cases: "higher" (#12) *vs.* "lower" (#2) impact velocity (Fig. 5) and confined (#14-1) vs. free (#14-4) shear impact experiment (Fig. 6).

#### 3.2.1. "Higher" vs. "lower" impact velocity

Fig. 5 shows that the general evolution of the pairs of stress intensity factors is rather similar, regardless of the impact velocity. The stress wave interacts with the crack-tip after some 10 μs while mode I is almost non-existent for the first 140 μs time after which it becomes markedly positive. Mode II is clearly dominant until about 140 μs.

- Specimen PC2 fractured by opening at a typical kink angle of 37°. Using a maximum hoop stress criterion, this value implies fracture during the mixed-mode loading phase, in the absence of a more accurate assessment of fracture time (Rittel and Maigre, 1996b; Rittel and Maigre, 1997). With the same criterion, failure during the initial mode II loading phase would yield a kink angle of about 70°.
- By contrast, specimen PC12 exhibited an adiabatic shear band which formed most likely during the initial mode II phase (first 140 μs).

One important difference between these two cases is in the different levels of mode II loading experienced by each sample. This observation can be related to shear band formation by suggesting that this failure mode is related to some critical (threshold) value of $K_{II}$. From Fig. 5 (and other



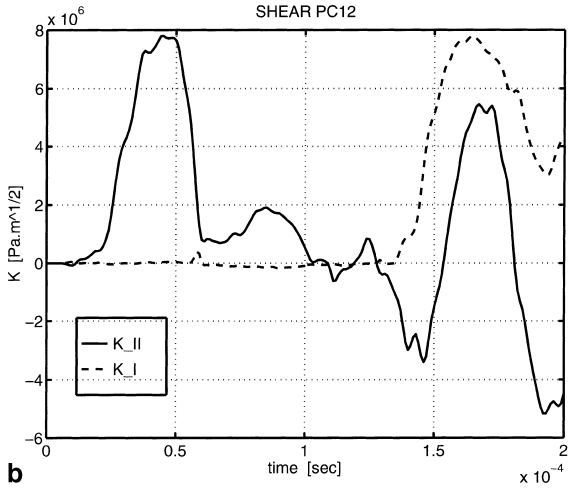
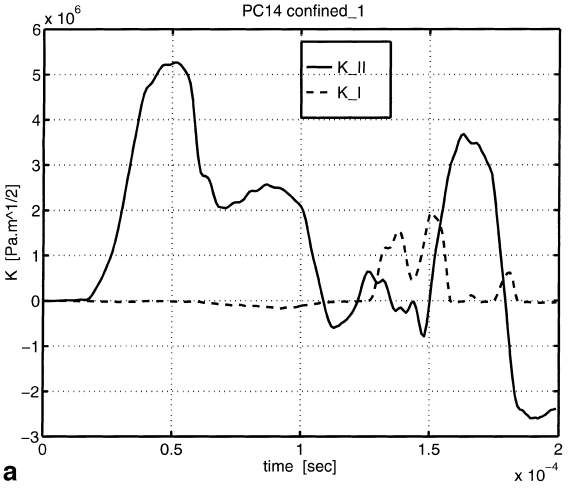




Fig. 5. Calculated evolution of the stress intensity factors for side impact experiments. (a) Specimen #2 – impact velocity 27 m/s. (b) Specimen #12 – impact velocity 60 m/s. Note the general similarity of the evolutions. Higher values are observed for higher impact velocities. Almost "pure mode II" is experienced for the first 140 μs followed by mixed-mode loading.



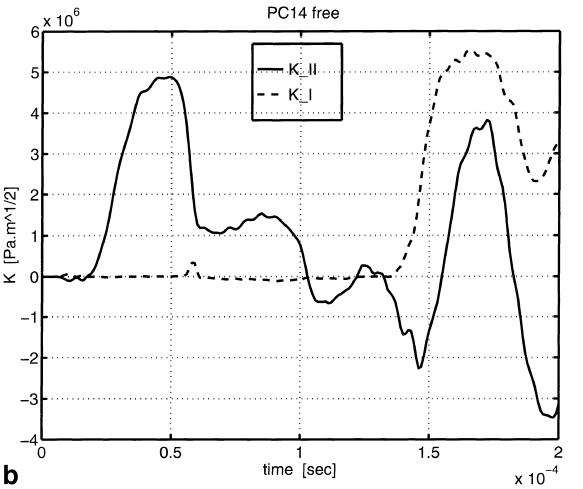




Fig. 6. Calculated evolution of the stress intensity factors for side impact experiments in specimen #14. (a) First shot with applied confinement. Impact velocity 32.5 m/s, specimen did not fracture but formed a shear band. (b) Fourth shot, free specimen. Impact velocity 32.5 m/s, specimen fractured.

similar results) a tentative value would be 4 MPa $m^{1/2}$ for this material. This is based on the observation that in Fig. 5(a) this value was not exceeded and kink cracking occurred by contrast with the results of Fig. 5(b) where $K_{II}$ exceeded 4 MPa $m^{1/2}$ and a shear band was observed. This value is selected as *the threshold below which no shear band was clearly observed*. However, it is a tentative value in the sense that it is determined from an assumed rather than measured value of the coefficient of friction in the numerical calculations. Yet it can serve as a useful parameter for future comparisons as shown in the sequel.

### 3.2.2. Confined impact

Specimen #14 was impacted four times in four distinct experiments. The first three times, confinement was applied whereas in the last shot confinement was released. In the first shot, a small

shear band formed which extended apparently very little during the two following shots. The last shot caused kinked fracture. The evolutions of the stress intensity factors for the first and the last shot of this specimen are shown in Fig. 6. It is first noted that for the constrained case, mode I is significantly reduced whereas for the same free specimen, mode I develops in a way similar to that shown in Fig. 5.

- This experiment confirms the suggestion that shear band nucleation is related to the sole evolution of $K_{II}$, as expected.
- It also shows that the nucleation (and limited growth) of the shear band and the subsequent fracture at a specific angle are two distinct events. Whereas the former is related to the evolution of $K_{II}$, the latter obviously depends on the evolution of $K_I$. This implies that each mechanism is triggered in a different range of times: shear band at earlier times, kink cracking at later times.

### 3.3. Dominant mode I (CCS)

The experimental results have been gathered in Table 3 for a total of 8 specimens. Here, fracture occurred in every case. However, similar to the shear impact experiments, the maximum kink angle does not exceed 40° regardless of the impact velocity (except for specimen 17 which may not be considered as typical). Yet additional information is available in these specimens, i.e. the fracture time measured from the fracture gages. However, it must be noted that this indication is the "upper bound" value of fracture time as it is measured on the surface of the specimen (for a discussion on this subject, see also Maigre and Rittel, 1996). It can also be noted that fracture is detected at earlier times for higher impact velocities.

### 3.4. Numerical simulation of selected compact compression experiments

Typical evolutions of the stress intensity factors are shown in Fig. 7 for lower (#14) and higher (#15) impact velocities. Such evolutions are typical of mixed-mode loading. The stress wave starts to interact with the crack-tip after about 40 μs. The graphs show that for the slower case, fracture occurs in the dominant $K_I$ phase, whereas for the higher impact velocity fracture is detected at an earlier phase where both modes are equally contributing. In this case too, the overall evolutions look very similar pairwise while once again the absolute value of the stress intensity factors increases with the impact velocity. Despite the different mixity ratios measured at fracture time, the fracture angles of the two specimens are very similar.

The evolutions of the shear $\sigma_{12}$ and hoop $\sigma_{\Theta\Theta}$ stresses in the vicinity of the crack-tip of specimen #15 (which fractured at $t = 78.5$ μs) are plotted in Figs. 8 and 9, respectively (the calculations have been carried out beyond actual fracture time to get a broader picture of the stress distribution). Past the initial 40 μs, shear stresses develop symmetrically with respect to the crack line at $t = 50$ μs and the symmetry is gradually lost gradually towards

Table 3
Summary of the experimental results for fatigue precracked compact compression specimens

| Sample # | $v_{imp}$ (m/s) | Crack ($\times 10^{-3}$m) | $t$-fract (μs) | Kink angle (°) | Remarks |
|---|---|---|---|---|---|
| 10 | 17.5 | 16.3–15.1 | n.a. | 42.5–31.5 | |
| 12 | 22.5 | 14.5–15.1 | n.a. | 32.0–33.0 | |
| 13 | 60 | 14.6–15.7 | n.a. | 34.5–40.0 | FEM |
| 14 | 22 | 14.4–15.3 | 115 | 27.0–32.0 | FEM |
| 15 | 45 | 15.3–15.8 | 78.5 | 34.0–38.0 | FEM |
| 16 | 25 | 14.7–16.3 | 98 | 31.5–40.5 | |
| 17 | 55 | 14.5–16.0 | 105 | 26.0–75.0 | |
| 19 | 28 | 13.5–15.5 | n.a. | 25.0–34.0 | |

The table indicates impact velocity, crack length as measured on both sides of the specimen, fracture time and the measured kink angle. FEM stands for FE modeling of this experiment.

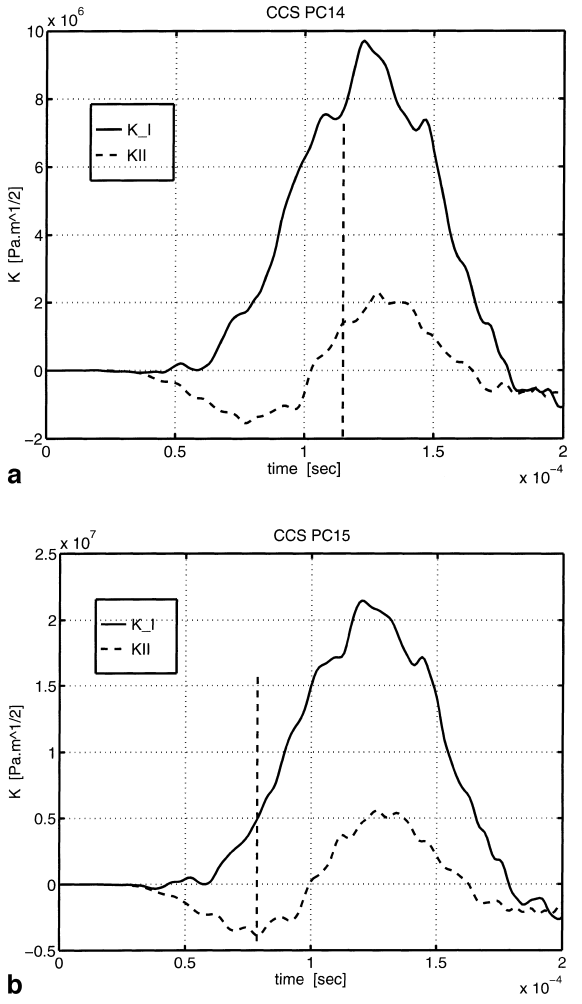



Fig. 7. Calculated evolution of the stress intensity factors for compact compression specimens. (a) Specimen #14 – impact velocity 22 m/s. (b) Specimen #15 – impact velocity 45 m/s. Note the general similarity of the evolutions. Higher values are observed for higher impact velocities. Mode I is dominant, but mixed mode is experienced during the whole experiment. Vertical dashed lines indicate fracture time.

$t = 95$ μs. At $t = 80$ μs (fracture time) large shear stresses are still experienced in a nearly symmetrical way over a typical distance of 0.5 mm ahead of the crack-tip. On the other hand, $\sigma_{\Theta\Theta}$ starts unsymmetrical at $t = 50$ μs and by $t = 95$ μs it has become totally symmetrical with respect to the crack-line. These evolutions are of course consistent with the evolutions of the corresponding stress intensity factors but they provide spatial information about the stress distribution.

### 3.5. Fractographic analysis

#### 3.5.1. Scanning electron microscopy

*3.5.1.1. Dominant mode II. "Higher" velocity:* The high velocity impact is characterized by a shear band which develops at the crack-tip but does not propagate significantly in our experiments, as shown in Fig. 4. The shear band appears as a planar zone, 1–2 mm long, which spans across the specimen's width (Rittel et al., 1997). This band is best detected by tilting the specimen with respect to the incident light and carefully looking in the vicinity of the crack-tip. Optical microscopy at relatively low magnifications discloses minor extension of the fatigue crack within the shear band but this crack did not propagate much either and its tip appears to be quite blunted (Fig. 10). Blunting can be qualitatively understood by assuming severe heating of the material in the band (Zhou et al., 1996a) which turns the medium into a viscous fluid. This point is further supported by the observation of tiny bubbles in the shear band area.

Fig. 11 shows the scanning electron fractograph of a confined specimen for which such a shear band was "grown" by repeated impact prior to final fracture following release of the confinement (specimen 23). The band is clearly distinguished with a typical pattern of parallel striations. The visible number of striations exceeds the number of repeated impacts so that the striated pattern cannot be assigned only to the repetitive nature of the loading. Rather, it indicates that failure within the shear band proceeds by discontinuous steps. The fractographic appearance is characterized by a "layered" aspect with some elongated voids visible at low magnifications (area 1 in Fig. 11). Examination of the band at higher magnifications reveals a lack of fine scale details. This fractograph is similar to that resulting from thermally induced damage on the surface of a PVC specimen (Engel et al., 1981). It may be taken as the indication of the high temperatures which develop in the shear band.

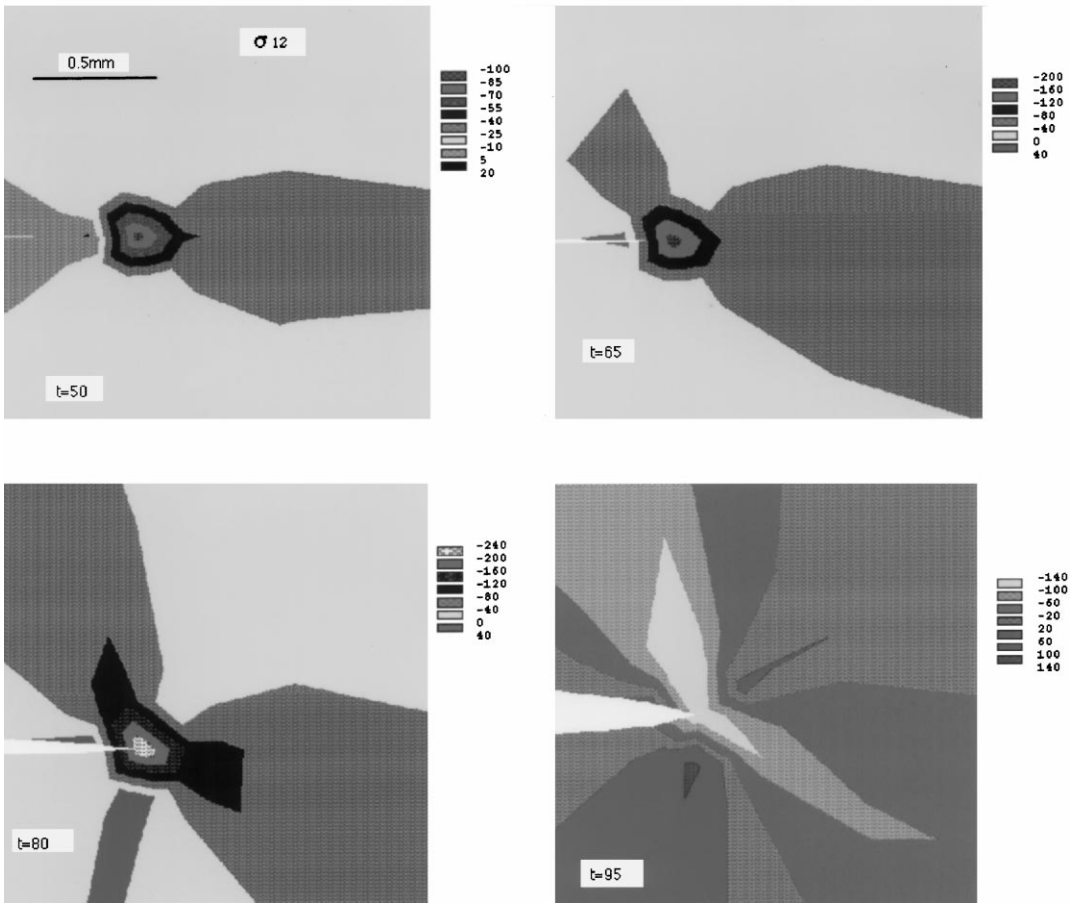



Fig. 8. Distribution of $\sigma_{12}$ (MPa) at selected times (μs) in the vicinity of the crack-tip of compact compression specimen #15 (fracture time $= 78.5$ μs).

The subsequent opening failure is characterized by a series of waves emanating concentrically from the shear band (area 2 in Fig. 11). Such wavy patterns are characteristic of impact loading in this material as a varying number of parabolic markings (waves) can be observed along the initial crack (shear band) front. The waves are emitted from a number of (mostly indiscernible) "sources" and they coalesce at a later stage of the crack initiation process to form the main crack front. Whereas such wavy patterns are not commonly reported for polymers, they characterize fracture of brittle materials, as shown on the macroscopic fracture surface of glass rods by Bodner (1973). Some wavy pattern can also be observed in the brittle fracture of PVC at −170°C (Engel et al., 1981). The overall size of the band and the waves is of the order of the millimeter.

The fracture surface topography in the waves appears to be featureless at low magnifications but inspection at higher magnifications reveals a characteristic fine scaled pattern of alternating patches of material over a uniform background. These patches can be attributed to the operation of a crazing mechanism sensitive to normal stresses (Kausch, 1987; Engel et al., 1981). Further apart (area 3 in Fig. 11) the fracture surface is much rougher. Yet, higher magnifications show again a lack of distinctive features.

"*Lower*" *velocity:* The fracture surface typical of a lower velocity impact – freely fractured specimen (#2, $v = 27$ m/s) is shown in Fig. 12. The

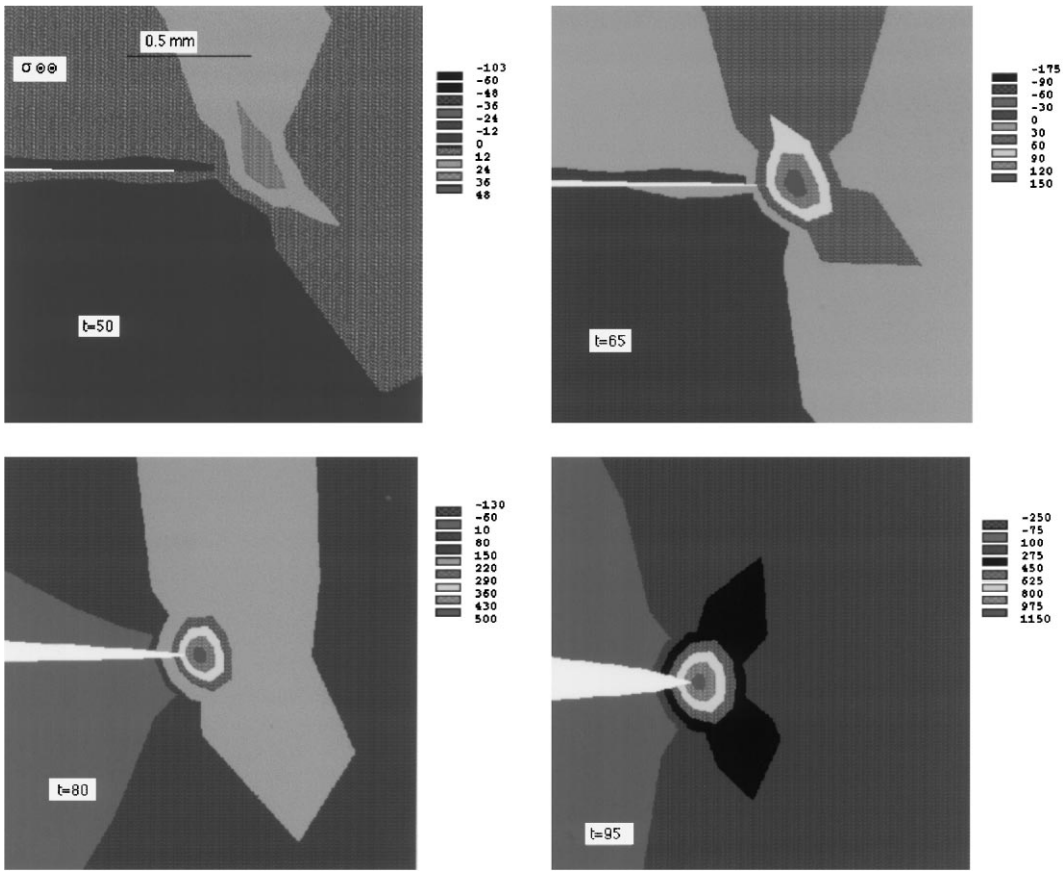



Fig. 9. Distribution of $\sigma_{\Theta\Theta}$ (MPa) at selected times (μs) in the vicinity of the crack-tip of compact compression specimen #15 (fracture time $= 78.5$ μs).

fracture comprises two main features: arrays of linear markings which converge to the fatigue crack. These arrays delineate a seemingly featureless zone (area 2 in Fig. 12), although patches of striations can somehow be discerned. At higher magnification, the trend is once again reverted! The linear features do not reveal fine scale details whereas in the smooth area regular patterns of striations are now clearly discernible. Such patterns were also observed by Ravi-Chandar (1995) who attributed them to the operation of a craze dominated fracture mechanism. Further examination at higher magnification clearly shows that the striations are made of alternating patches identical to those observed in the previous specimen.

*3.5.1.2. Dominant mode I. "Higher" velocity:* A typical fractograph of higher velocity impact in a compact compression specimen (#13) is shown in Fig. 13. The general aspect of the fracture surface looks quite familiar at this stage. Fracture begins by a shear band with its characteristic pattern of parallel striations. This pattern confirms the lack of correspondence between the number of striations and the number of impacts on the one hand as well as the discontinuous nature of the fracture process in the band. Higher magnifications reveal the characteristic layered aspect of the fracture surface in the band. The band extends by a series of waves within which one can observe the alternating patches of material, both features similar to

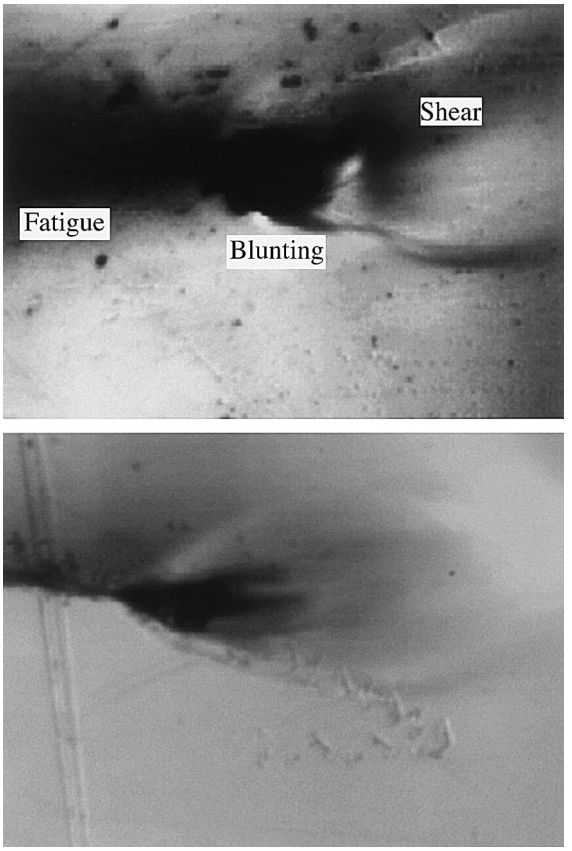



Fig. 10. Optical micrograph of arrested cracks in a shear band. Note the crack-tip blunting.

those observed in the side impact specimen. Farther away, the fracture surface is very rough, quite similar to the observations of the previous specimen.

"*Lower*" *velocity:* A typical fractograph of lower velocity impact in a compact compression specimen (#10) is shown in Fig. 14. In the immediate vicinity of the fatigue crack, filaments are noticeable (similar to the previously mentioned linear features) and the general fracture surface topography seems once again rather dull at low magnifications, whereas at higher magnifications, the characteristic alternating pattern of patches is again observed. Here too, the fracture surface topography comprises rough features farther away from the initiation area.

*3.5.2. Optical microscopy*

Fig. 15 is a side view of a low velocity impacted CCS (#10 – 17.5 m/s). The crack propagates at the reported angle with respect to the fatigue crack tip and no special features are noticeable.

Fig. 16 shows the same view at sensibly similar magnification of a specimen impacted at a much higher velocity (#13 – 60 m/s). The striking difference consists of two damage zones around the fatigue crack-tip. One is smaller, appears darker and is roughly parallel to the crack line. Its length size is equal to about 0.5 mm. This zone looks quite similar to the shear zone observed in the side impact experiments and predicted by the numerical simulations. This size corresponds to the characteristic size of a shear band observed in the fractographic analysis. Aside this zone is a white zone, about 1 mm wide which appears as a damage zone whose trace underlies the fracture path. This zone can tentatively be identified as the zone in which significant hoop stresses develop at fracture time. The opening fracture appears to initiate at an angle of 50° or above to the original crack over a very short segment within the shear band rather than at its tip. Past this initial segment, the opening crack adjusts itself to the typical angle values reported previously. This sequence of fracture events can be understood by recalling that fracture may initiate with limited propagation within the shear band as shown previously. It thus seems that the opening crack is the continuation of the previously arrested crack as dominantly opening conditions prevail at later times.

*3.5.3. Summary of the fractographic analysis*

To summarize the main result of the fractographic analysis, we observed that for comparable impact velocity (higher or lower) the fracture surfaces of our two types of specimens – which represent the two extremes of mode – mixity – *are virtually indistinguishable*. We identified the characteristic fracture micromechanisms representative of shear and opening failure. These failure modes are operative in general mixed-mode loading. As a result, shear failure can dictate the initial stages of fracture of a dominant mode I experiment for which the mode II, despite being the minor mode, is not negligible.

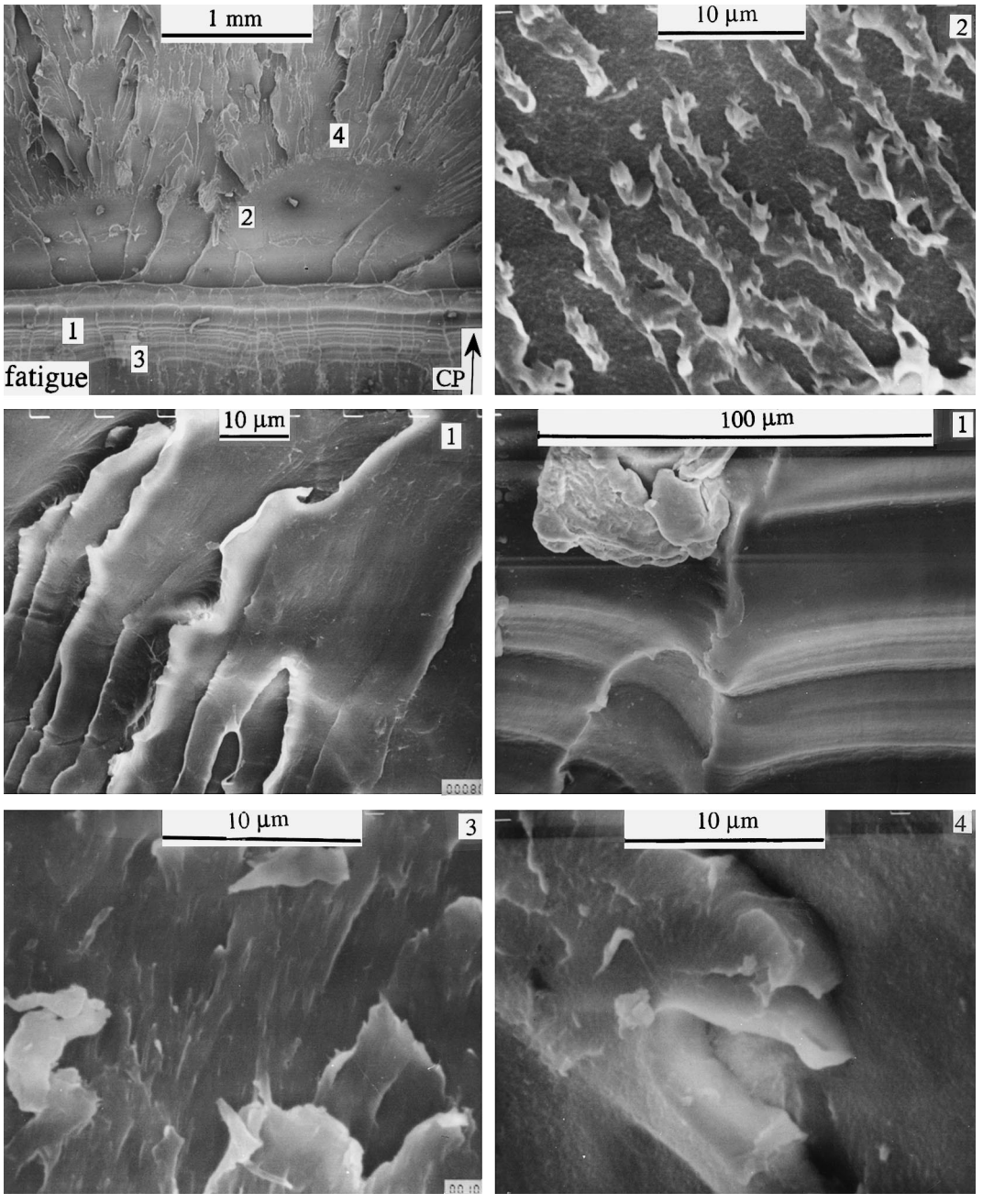



Fig. 11. Scanning electron fractograph of side impact specimen #23 – impact velocity 22.5–32.5 m/s. Repeated impacts. CP indicates crack propagation direction. A general view of the initiation area is shown in the upper left photograph. (1) is the shear band area, (2) indicates the wave region with craze fracture, (3) is the boundary between the fatigue crack and the shear band and (4) is the rough area adjacent to (2).

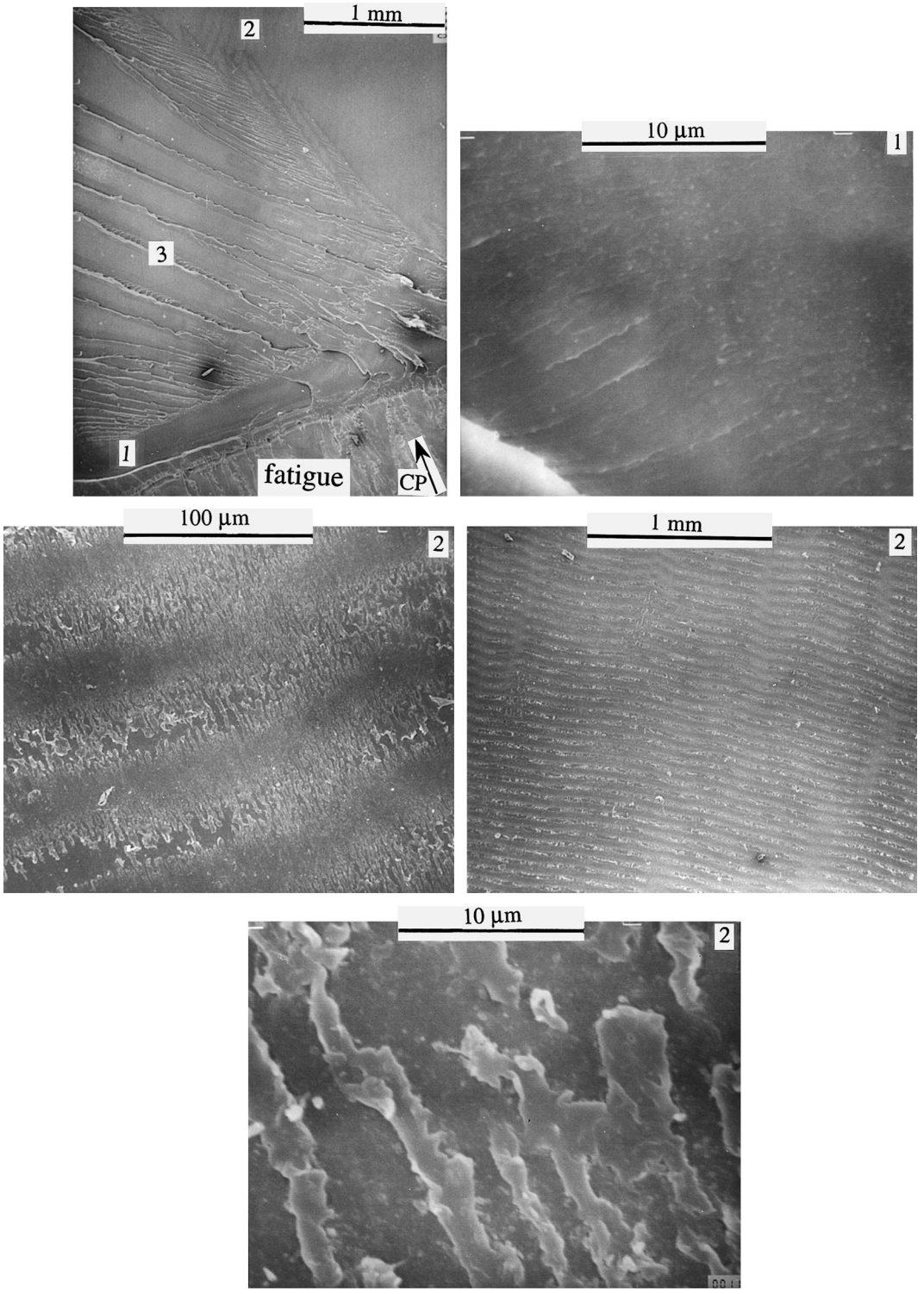



Fig. 12. Scanning electron fractograph of side impact specimen #2 – impact velocity 27 m/s. A general view of the initiation area is shown in the upper photograph. (1) is the boundary between the fatigue and the fast crack. Filaments are observable in side areas (3). Central zone (2) comprises regular patterns indicative of craze dominated fracture, shown at various magnifications.

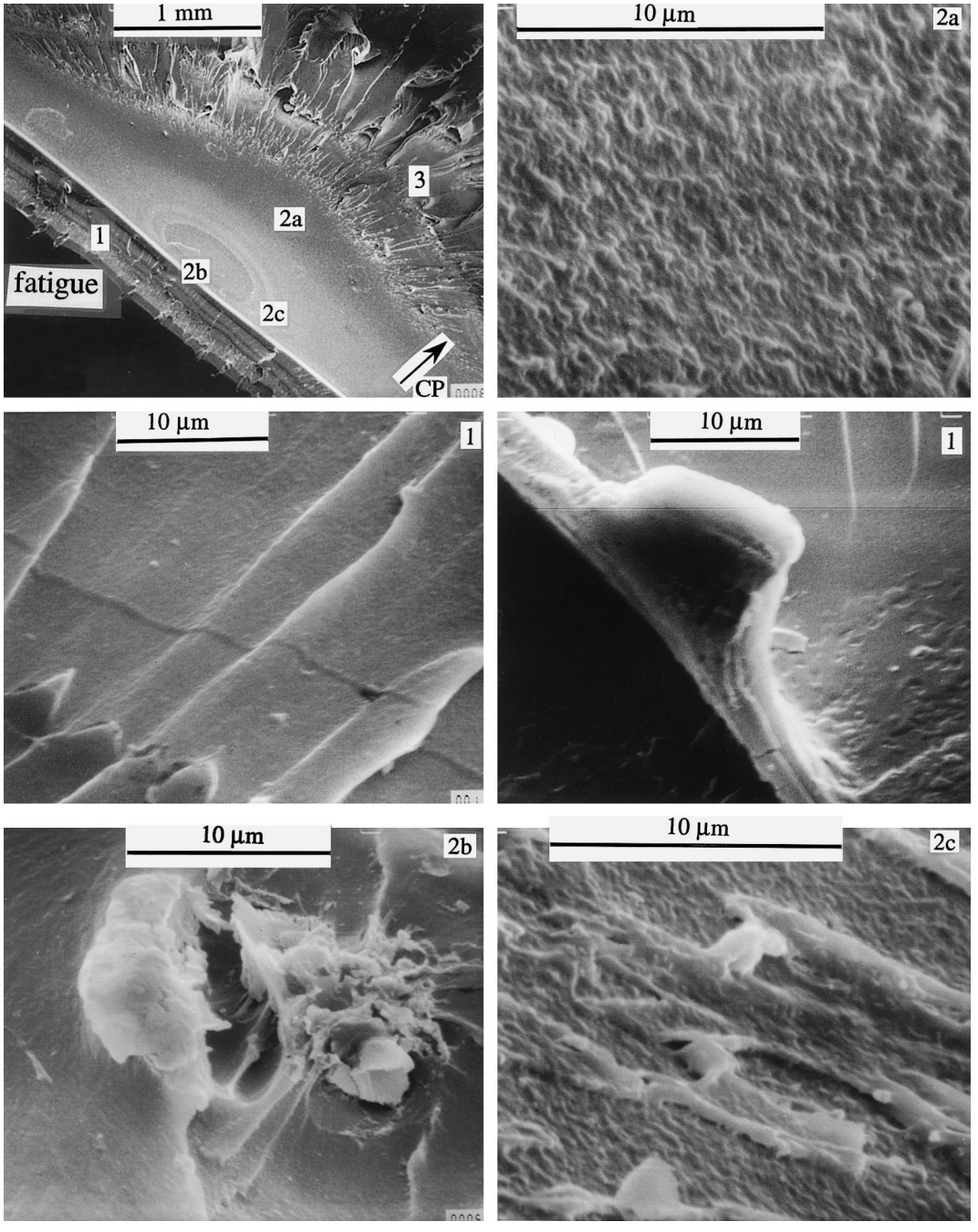



Fig. 13. Scanning electron fractograph of compact compression specimen #13 – impact velocity 60 m/s. Note the general analogy with Fig. 11. A general view of the initiation area is shown in the upper left photograph. (1) is the shear band area 2a and 2b indicates the wave region with craze fracture, (3) is the rough area adjacent to (2).

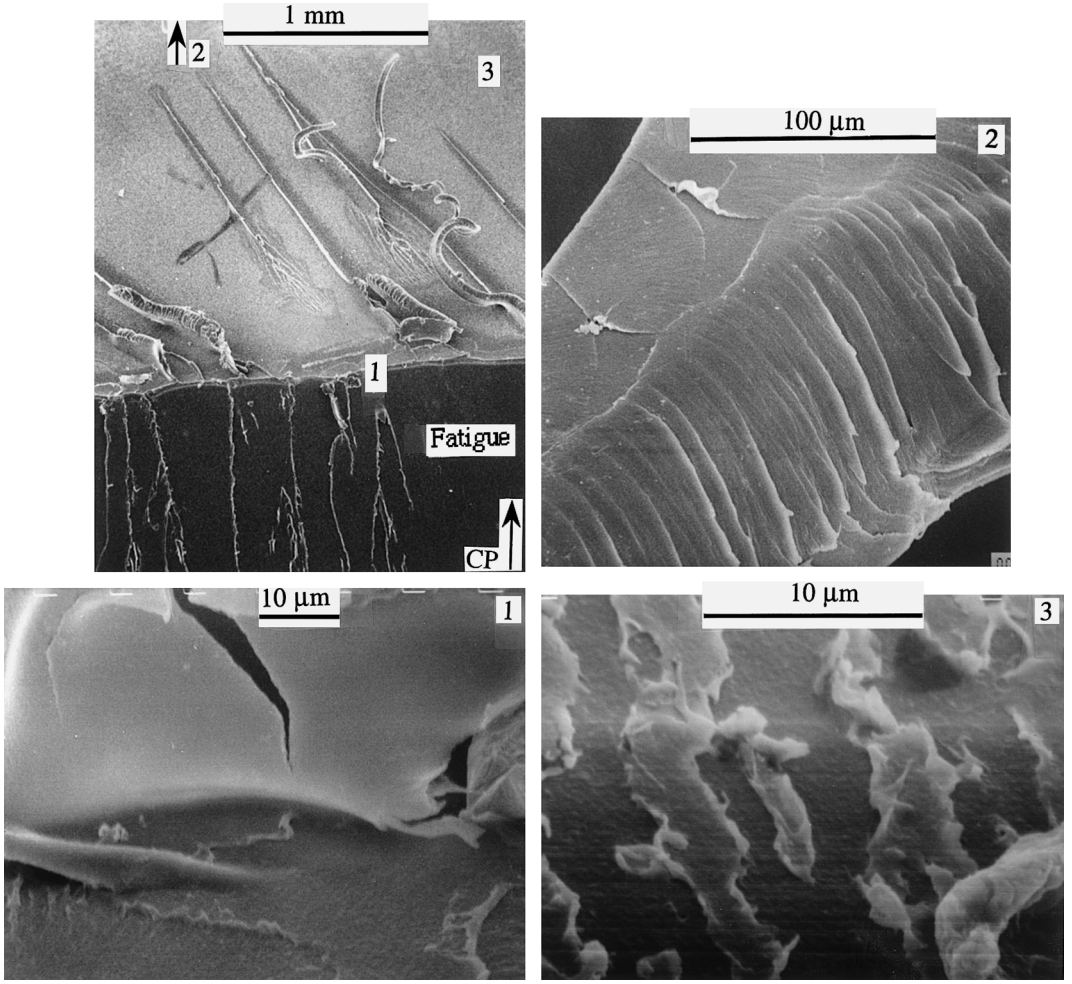



Fig. 14. Scanning electron fractograph of compact compression specimen #10 – impact velocity 17.5 m/s. Note the general analogy with Fig. 12. The crack initiation area comprises filaments (1) and looks featureless at low magnifications. Higher magnification shows the operation of craze dominated fracture. Farther away, the fracture surface comprises rough features.

## 4. Discussion

- The investigation has been carried out along three complementary axes: mechanical (dynamic fracture tests) – materials related (fracture mechanisms) – and numerical.
- A relatively large sample size of dominant mode II and dominant mode I specimens have been systematically tested and compared using the framework of LEFM.
- The experiments reported here all involve mixed-mode loading. While confirming the previously reported trend for failure mode transition he present results shed additional light on its operation in the more general context of mixed-mode loading.
- A first observation is that transition of failure modes is not related to the nature of the crack-tip (notch or fatigue crack). The selection of a failure mode is thus not sensitive to the level of negative mode I which develops in side impact experiments.
- A key point in our mode II experiments is that the crack-tip experiences initially almost pure

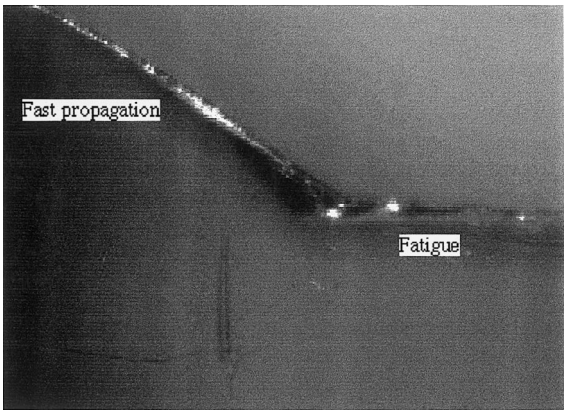



Fig. 15. Side view of compact compression specimen #10. The crack propagates initially at 42°–31.5° (fractograph shown in Fig. 14).

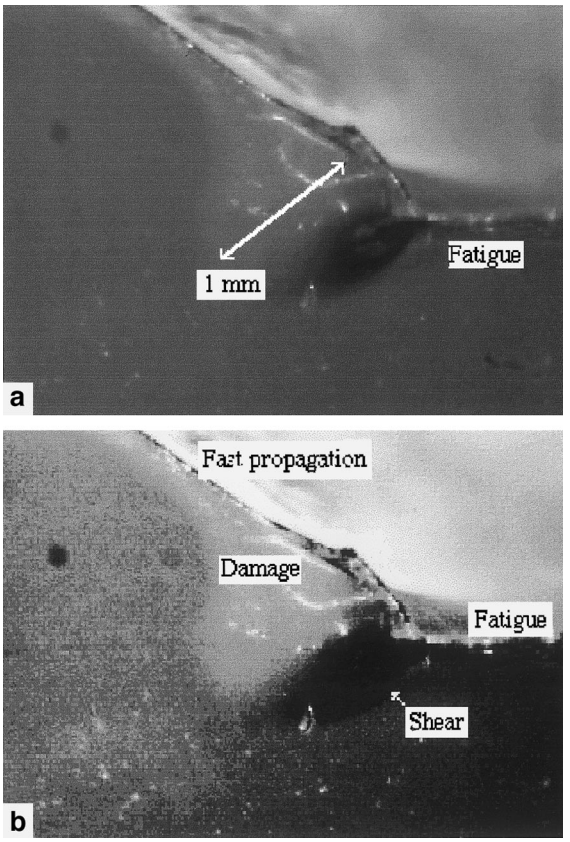



Fig. 16. Side view of compact compression specimen #13. The crack propagates initially at an angle superior to 50° followed immediately by an angle of 34.5–40° (fractograph shown in Fig. 13). (a) As recorded. (b) After image processing. Note the dark shear area, about 0.5 mm long immediately ahead of the crack-tip. The white area is about 1mm wide and results probably from the operation of normal stresses. Note the wide trace underlying the fast fracture path.

mode II conditions followed by mixed-mode loading at the later stages. Shear band initiation, propagation and final fracture are thus *separable events* in this experiment.

The *initiation* phase of dynamic cracking with the occurrence (or lack) of adiabatic shear banding is the characteristic response of a material which possesses *a sensitivity to two distinct failure modes*, shearing and opening. As mentioned previously, the same experiment carried out on a brittle polymer – characterized by a single failure mode (opening) – always yields fracture at about 70°. Consequently, the local response of the crack-tip material to the global $K$ dominated fields is dictated by its *failure criterion*. In other words, the same crack-tip material may fail according to a maximum normal stress criterion in its isothermal state while it may fail according to a maximum shear stress criterion when significantly heated. A threshold value for $K_{\mathrm{IISB}}$ (4 MPa $\mathrm{m}^{1/2}$) was tentatively postulated as a criterion for shear band nucleation in the heated polycarbonate. This criterion is attractive in the sense that it may serve as a basis for comparing various materials and experiments from different sources.

Considering now *crack propagation and final fracture*, the series of confined experiments show that one can trigger and even ''grow'' adiabatic shear bands by repeated impact. However, final fracture of our polycarbonate specimens always involved local opening (mode I) which depends on the confinement applied. This point is related to the very limited shear band propagation in our experiments. We also observed limited crack growth within the shear band, as well as its significant blunting which apparently results from local softening of the surrounding material. When other materials (especially met-

als) were studied, shear band propagation over noticeable distances was reported, including eventual specimen separation. To conciliate these observations about dynamic crack *propagation* one must recall the results of Zhou et. al. (1996a, b) who reported that the shear band propagation comes to a stop when the shear loading drops until subsequent wave reflection. Shear band propagation and ensuing specimen separation are thus highly dependent on the specific experiment carried out and investigated material's response, as shown both numerically and experimentally. We observed that fracture proceeds as the extension of the arrested crack within the band rather than the opening of the shear band's tip.

- We observed that occurrence of shear bands is neither restricted to a specific kind of experiments nor to the crack-tip geometry. Rather, it is related to the *mode-mixity* of the loading. This point was is further corroborated by the striking similarity of fractographic features between mode II and mode I specimens at matching impact velocities. As expected, the impact velocity leaves a characteristic "signature" on the fracture surface.

## 5. Conclusion

Dynamic crack initiation has been *systematically* studied in fatigue precracked polycarbonate specimens using two different series of experiments characterized by *mixed-mode loading*. A simple hybrid experimental–numerical framework was used to analyze and compare a large number of experiments in two setups which cause dominant mode I or mode II loading, respectively. The following conclusions can be drawn from this study.

- The experiments confirm observations of failure mode transitions and adiabatic shear band formation in notched/cracked plates.
- The development of negative mode I is prevented in sharp fatigue cracks. Our numerical simulations show that such specimens experience almost pure mode II loading conditions for a large part of the experiment.
- Shear bands can nucleate at the tip of fatigue cracks independently of the level of negative mode I.

In addition, our results also show that *for different mode-mix experiments*:

- Shear bands develop when $K_{II}$ exceeds a postulated threshold value (tentatively fixed to 4 MPa $m^{1/2}$), which describes the adiabatically heated crack-tip material.
- The fractographic study allowed identification of specific fracture mechanisms which characterize with little ambiguity the applied impact velocity.
- Provided significant (adiabatic) crack-tip heating develops, shear failure may be observed to operate in general mixed-mode loading.
- As such, the operation of a shear failure mode is not restricted to loading configurations in which mode II dominates.

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